NoSQL Data Model

Stripe collects a number of unstructured data points that cannot be stored in relational databases like OLTP or OLAP. This includes, for example, user interaction with the website, error reports, data used to train the machine learning model, and user feedback. Therefore, we created a NoSQL database to store this data. This database requires specific strategies to optimize its speed and integrate with OLTP and OLAP databases.

1. Strategy for Managing Relationships Between Documents

The first strategy is to create an embedding for data that is unlikely to change. For this reason, it was implemented for data related to user interaction with the application. Each click and navigation by the user was thus directly embedded in the document. This allows for quick reading, as all the data concerning this behavior is in one place.

The second strategy involves referencing, meaning that documents are linked to existing data. This was done for all identifiers, such as customer_id, transaction_id, and merchant_id. This ensures data consistency and facilitates updates.

Finally, indexing was implemented to quickly retrieve specific information. This is why this technique should be used for information that is frequently accessed. It was therefore applied to errors, particularly those in transaction identifiers such as transaction.transaction_id or transaction.customer_id. This allows for quick error retrieval rather than having the software scan every row in the database to find errors related to a merchant or user.

2. NoSQL Integration with OLTP and OLAP Systems

This NoSQL architecture integrates with OLTP and OLAP architectures in several ways.

First, the presence of the identifiers mentioned above allows for linking the OLTP architecture with NoSQL. For example, when a transaction occurs in the OLTP system, the transaction_id is received in the NoSQL system. This same identifier is then stored in MongoDB.

Second, the NoSQL system feeds certain information into the OLAP architecture. This is the case for fraud detection, which requires predictions from the machine learning model fed by the NoSQL document. These inputs enable the prediction of a fraud score, which enriches the OLAP database.

To function, the system is orchestrated by two software packages: Kafka and Apache Airflow. Kafka handles real-time data streaming. It allows for the collection of events in the OLTP database and the distribution of information to OLAP and NoSQL systems. Airflow acts as an orchestrator, regularly extracting data from the OLTP system to the OLAP system and synchronizing NoSQL data with OLAP. The two work together to enable the transition of all data types.

The NoSQL system presented here therefore has three methods for optimizing the computing power of the NoSQL database. Embedding allows for the rapid retrieval of certain information that is unlikely to change over time. Referencing maintains consistency between databases and facilitates updates. Finally, indexing allows for the rapid manipulation of certain data. The system can thus gain speed with better resource management. Furthermore, it is designed to work in conjunction with other types of OLTP and OLAP architectures.